# Newton's Laws and Friction

Physics 101  Your Name

## 1 Newton's second law

The net force on an object equals its mass times its acceleration:

$$\vec{F}_{\text{net}} = m\,\vec{a}. \tag{1}$$

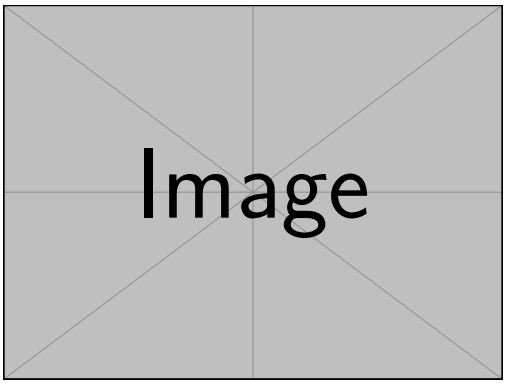
Equation (1) holds in an inertial reference frame.

## 2 Friction

### 2.1 Static and kinetic friction

Use real list environments for lists:

- Static friction satisfies $f_s \leq \mu_s N$.
- Kinetic friction satisfies $f_k = \mu_k N$.




| Surfaces | $\mu_s$ | $\mu_k$ |
|---|---|---|
| Rubber on dry concrete | 1.0 | 0.8 |
| Steel on steel | 0.74 | 0.57 |